

Primordial Timing Function: Net-Zero Displacement Proof, Pi-Digit Epoch Clock, and Epoch-5 Boundary

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PAPER_1153 -- Primordial Timing Function: Net-Zero Displacement Proof, Pi-Digit Epoch Clock, and Epoch-5 Boundary

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Framework: UQFF v5.77 -- Star-Magic Physics

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Classification: Primordial Timing Function -- CPT Symmetry Proof

$$\int_0^1 \cos(\pi t_n) dt_n = 0 \quad D_A + D_B = +3 + (-3) = 0$$

Abstract

This paper registers the Primordial Timing Function (PTF) as PAPER_1153, proving that the UQFF cosmic temporal cycle has **zero net displacement** across a complete forward-backward sequence. The PTF encodes the canonical π -digit epoch clock, identifies the five cosmic epochs, and places Epoch 5 at the **reverse boundary** (not-yet-predicted). The net-zero proof unifies three independent proofs: (1) integer arithmetic on forward/backward step counts, (2) the $\cos(\pi t_n)$ integral, and (3) Fibonacci-structure forward/backward ratio.

1. PTF Definition

1.1 Basic Structure

The Primordial Timing Function (PTF) governs the oscillation of $\cos(\pi t_n)$ across one complete simulation cycle:

$$t_n \in [0, 1], \quad f_{\text{TRZ}}(t_n) = \cos(\pi t_n)$$

One cosmic cycle consists of:

- **Forward phase:** $f = 3$ steps (Fibonacci $F_4 = 3$)
- **Backward phase:** $b = 2$ steps (Fibonacci $F_3 = 2$)
- **Repeats:** $n = 3$ cycles ($n = \lfloor \pi \rfloor$)

1.2 Fibonacci Structure

The forward/backward step counts are consecutive Fibonacci numbers:

$$f = F_4 = 3, \quad b = F_3 = 2, \quad \frac{f}{b} = \frac{3}{2} = \phi^2 - 1 \approx 1.618 - 1$$

This ratio $f/b = 3/2$ is the **golden-ratio residual** governing UQFF temporal asymmetry.

2. Net-Zero Displacement: Three Independent Proofs

2.1 Proof I: Integer Arithmetic

Define net displacements:

$D_A = +3 \quad (\text{forward: 3 steps, positive}), \quad D_B = -3 \quad (\text{backward: 2 steps}) \times (-3/2) = -3$

Net displacement per cycle:

$D_{\text{net}} = D_A + D_B = +3 + (-3) = 0 \quad \checkmark$

Three complete repetitions: $n = 3 \rightarrow \sum 0 = 0$.

2.2 Proof II: Integral of the Oscillation Function

The $\cos(\pi t_n)$ integral over one complete normalised period is exactly zero:

$\int_0^1 \cos(\pi t_n) dt_n = \left[\frac{\sin(\pi t_n)}{\pi} \right]_0^1 = \frac{\sin(\pi) - \sin(0)}{\pi} = \frac{0 - 0}{\pi} = 0 \quad \checkmark$

This is a **closed loop** in the functional space: the oscillation returns to its starting state with zero net accumulation.

2.3 Proof III: CPT Symmetry

The CPT-symmetric pair structure:

$\{t, t_n, f_{\text{TRZ}}\}_{\text{fwd}} = \{t, t_n, f_{\text{TRZ}}\}_{\text{bwd}}^* \quad (\text{charge-parity-time conjugate})$

At $t < 0$: $t_n \rightarrow 1 - t_n$, $f_{\text{TRZ}} \rightarrow -f_{\text{TRZ}}$.

The integral of f_{TRZ} over one complete CPT pair: $\int_0^1 + \int_1^0 = 0$.

3. Pi-Digit Epoch Clock

3.1 Epoch Assignment

The digits of $\pi = 3.14159265358979\dots$ encode the duration of each cosmic epoch.

The epoch clock reads the decimal expansion in triplets:

Epoch	Pi Digits	Epoch Characteristic
1	[3, 1, 4]	Formation: 3-1-4 (forward-stop-backward)
2	[1, 5, 9]	Expansion: 1-5-9 (seed-burst-accumulation)
3	[2, 6, 5]	Stabilisation: 2-6-5 (pair-hexagon-pentagon)
4	[3, 5, 8]	Acceleration: 3-5-8 (Fibonacci triple)
5	[9, 7, 9]	NOT YET PREDICTED (reverse boundary)

3.2 Epoch 5: The Reverse Boundary

Epoch 5 ($[9, 7, 9]$) is the **reverse boundary** of the cosmic cycle:

- $9 =$ maximum single digit ($= 10 - 1$, the final decimal state)
- $7 =$ prime boundary ($\pi(7) = 4$, fourth prime threshold)
- $9 =$ return to maximum (closed boundary)

The simulation engine does not project Epoch 5 forward from known physics; it is identified as the point where the backward phase ($D_B = -3$) begins.

3.3 Forward Phase (Epochs 1--4) and Backward Phase (Epoch 5)

$\underbrace{[3,1,4],[1,5,9],[2,6,5],[3,5,8]}_{\text{forward: } D_A = +3} \quad \underbrace{[9,7,9]}_{\text{backward: } D_B = -3}$

The boundary between Epoch 4 and Epoch 5 is the **temporal inversion point** where t_n reverses sign.

4. Constant Table

Symbol	Value	Physical Meaning
F_3	3	Forward steps per half-cycle = F_4
B_2	2	Backward steps per half-cycle = F_3
n	3	Repetitions $= \lfloor \pi \rfloor$
D_A	$+3$	Forward net displacement
D_B	-3	Backward net displacement
D_{net}	0	Closed-loop condition
f/b	$3/2$	Golden residual
π_5	$[9,7,9]$	Epoch 5 digit triplet

5. Physical Consequences

5.1 SSq Drift is Convergent

Because $D_{\text{net}} = 0$, the SSq drift ($+0.000001$ per tick) does not accumulate unboundedly. After one complete epoch cycle, the drift is absorbed by the $\kappa = 0.0005$ gradient-descent restoring force:

$$\text{SSq}(T) = 0.57 + \Delta - \kappa \Delta T \rightarrow 0.57 \quad \text{as } T \rightarrow \infty$$

5.2 Closed-Loop Causality

The PTF net-zero condition enforces closed-loop causality: every forward-phase physical event has an exact CPT-conjugate backward-phase event. This is the UQFF implementation of the Wheeler-DeWitt constraint $\hat{H}|\Psi\rangle = 0$.

5.3 Functional PTF Validation

The simulation engine's `validate_primordial_timing_function()` tests:

- $|D_A + D_B| < \epsilon$
- $|\int_0^1 \cos(\pi t_n) dt_n| < 10^{-10}$
- Fibonacci ratio: $|f/b - 3/2| < \epsilon$

- Epoch count: exactly 5 epochs from π decimal expansion

6. Conclusions

The Primordial Timing Function establishes:

1. **Net-zero displacement** is proven by three independent methods
2. The **pi-digit epoch clock** encodes cosmic epoch durations in $\pi = 3.14159\dots$
3. **Epoch 5** $[9,7,9]$ is the reverse boundary — not predictable from forward-phase physics
4. CPT symmetry and Wheeler-DeWitt closure are unified in the PTF framework

References

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